Auto-Loan Exercise: A Sequence of Utility Specifications and the Choice of Decision Period

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Discussion draft for Eddie

Abstract

This note answers the two items from our last discussion. Section 2 writes down eight utility specifications, from a static choice with at most one car to a dynamic model with latent need. Each one states the state, the menu, the utility, the choice probabilities, the forecast, what it adds to the previous version, and what it still cannot do. Section 3 discusses the decision period. Section 4 lists the choices made so far and the ones I would like to settle with you. No parameters are estimated, and identification is not yet shown for any version.

1 Common setup

Notation. Household i, month t. At the start of month t the household is in state x_{it}. The state summarizes everything about the household that matters for the decision, and **its dimension is fixed within a version**. The household picks one action d per month from a menu $\mathcal{M}(x)$. The menu depends on the state but never grows over time. One transaction per month at most.

States and actions. The number of cars held, n, is part of the state. "No car" is a state, not an action, and so is need. The actions are:

Action	Available when	Counts as a purchase?
N: do nothing	always	no
B: buy a first car	$n = 0$	yes
R: replace a car (buy one, dispose of one)	$n \geq 1$	yes
S: sell or dispose of a car without replacing it	$n \geq 1$	no
A: buy an additional car	$1 \leq n < \bar{n}$	yes

The maximum number of cars $\bar{n}$ is raised one step at a time. Versions 1–6 use $\bar{n} = 1$, so $n \in \{0, 1\}$ and A is never available. Version 7 raises the cap to $\bar{n} = 2$. A purchase is any B, R or A. This matches the event in the proposal, which counts first purchases, additions and replacements.

Post-action fleet. Throughout, a is the number of months the household has held its current car, not the vehicle's age since manufacture. A newly acquired car, new or used, has $a = 0$; the difference between new and used cars enters through the tier in Version 5. Write n^d and a^d for the number of cars and the holding time of the car held right after action d:

- N leaves both unchanged;
- B and R give $n^d = 1$ and $a^d = 0$;

- S gives $n^d = 0$.

With no car, the holding time is coded as empty, so the state keeps its dimension. One month later, a household that still has a car has $a' = \min(a^d + 1, \bar{A})$, where $\bar{A}$ caps the holding time to keep the state space finite.

Shocks. Each available action receives an additive shock $\varepsilon_{it}(d)$, i.i.d. type-I extreme value with unit scale, independent of x_{it}. The household chooses the action with the largest $U(x_{it}, d) + \varepsilon_{it}(d)$. Its choice probabilities are therefore

$$\mathbb{P}(d \mid x) = \frac{\exp U(x, d)}{\sum_{b \in \mathcal{M}(x)} \exp U(x, b)}. \tag{1}$$

In the dynamic versions, U is replaced by the choice-specific value v.

Target. From the forecast origin (end of month t), the probability of at least one purchase in the next H months:

$$F_i(t, H) = \mathbb{P}\big(\text{purchase in } t+1, \ldots, t+H \mid \mathcal{I}_{it}\big), \qquad H \in \{6, 12\}. \tag{2}$$

Forecasting needs a state law in every version. Even a static model needs to know how the state moves in order to forecast beyond one month. Let $\mathcal{N}(x)$ be the non-purchase actions at x: $\{\mathsf{N}\}$ when $n = 0$, and $\{\mathsf{N}, \mathsf{S}\}$ when $n = 1$. Let $T_d(x' \mid x)$ be the transition after action d. The state at the start of month $t+1$ is taken as observed at the forecast origin, since the end of month t and the start of month $t+1$ are the same point. Starting from $\mu_0 = \delta_{x_{i,t+1}}$, define the mass of households with no purchase so far:

$$\mu_{k+1}(x') = \sum_x \mu_k(x) \sum_{d \in \mathcal{N}(x)} \mathbb{P}(d \mid x)\, T_d(x' \mid x), \qquad F_i(t, H) = 1 - \sum_x \mu_H(x). \tag{3}$$

A household that sells its car has not purchased, but it moves to $n = 0$, where it may later buy. So (3) needs the transitions after N and S. Every version below uses (3); only the choice probabilities, the non-purchase set and the state change.

One flow utility for the whole sequence. Every version is built on the same flow utility. The flow is the value of the fleet held after the action, minus the cost of the transaction:

$$u(x, d) = n^d\big(g - \eta\, a^d\big) - RC\, \mathbf{1}\{d \in \{\mathsf{B}, \mathsf{R}\}\} - \kappa_S\, \mathbf{1}\{d = \mathsf{S}\}. \tag{4}$$

The parameters are:

- $g > 0$, the monthly value of having a car rather than none;
- $\eta > 0$, the monthly cost of each additional month of holding (maintenance and depreciation);
- RC, the net cost of a purchase;
- κ_S, the net cost of selling (hassle minus proceeds; its sign is not restricted).

Doing nothing with no car gives zero. That is a normalization, and only differences in utility matter.

Reading (4) by state:

- at $n = 0$: $u(\mathsf{N}) = 0$ and $u(\mathsf{B}) = g - RC$;
- at $n = 1$: $u(\mathsf{N}) = g - \eta a$, $u(\mathsf{R}) = g - RC$, and $u(\mathsf{S}) = -\kappa_S$.

2 The sequence of specifications

Table 1 summarizes the sequence. Each row adds one thing to the version it builds on; the sequence branches after Version 4, as explained below the table. The last column says which part of the Need–Ability–Choice structure from the review it addresses.

k	State x_{it}	Menu at n=0 / n=1	dim	Added, and why	Layer
1	(n, a)	$\{\mathsf{N}, \mathsf{B}\}$ / $\{\mathsf{N}, \mathsf{R}, \mathsf{S}\}$	2	Baseline, static: purchases depend on whether a car is held and how long	Choice
2	(n, a, ℓ)	same	3	Buying is harder when liquidity is low	Ability
3	(n, a, ℓ)	same	3	Forward-looking: value of waiting	Choice
4	(n, a, ℓ); type τ	same	3 (+type)	Persistent differences between households	Choice
5	(n, a, q, ℓ)	tiers of B, R	4	Which car: used or new	Choice
6	(n, a, q, ℓ, b)	cash or loan for B, R	5	Financing: the auto-loan event itself	Ability
7	(n, a_1, a_2, ℓ)	adds A; $n \leq 2$	4	Second cars	Need, Choice
8	base + latent z	as in base	base (+1)	Life events change the state, not the menu	Need

Table 1: The proposed sequence. a is months the current car has been held, ℓ liquidity level, q the tier of the car held, b months left on an auto loan, τ a persistent type, z latent need. Versions 1–6 cap the number of cars at one; Version 7 raises the cap to two. The sequence branches after Version 4: Versions 5–6 add which car and financing, and Version 7 adds second cars, each starting from Version 4.

Versions 1–4 form a single line. After that the sequence branches: Versions 5 and 6 build on Version 4, and Version 7 also builds on Version 4, not on Version 6. Within a branch, each version keeps everything in the versions above it unless it says otherwise. In particular, every version from Version 3 on is dynamic: choice probabilities are (1) with the values v of (6)–(7), computed with that version's flow utility and transitions.

Version 1: static, at most one car

State $x = (n, a)$, with $n \in \{0, 1\}$ and $a \in \{0, \ldots, \bar{A}\}$ (empty when $n = 0$). **Menu** $\{\mathsf{N}, \mathsf{B}\}$ at $n = 0$ and $\{\mathsf{N}, \mathsf{R}, \mathsf{S}\}$ at $n = 1$. **Utility** $U_1(x, d) = u(x, d)$ from (4).

Choice probabilities. Binary logit at $n = 0$: $\mathbb{P}(\mathsf{B} \mid 0) = \Lambda(g - RC)$, with $\Lambda(z) = 1/(1 + e^{-z})$. Three-way logit (1) at $n = 1$, where, relative to doing nothing,

$$U_1(\mathsf{R}) - U_1(\mathsf{N}) = \eta a - RC, \qquad U_1(\mathsf{S}) - U_1(\mathsf{N}) = -(g - \eta a) - \kappa_S.$$

Static means the household compares only this month's payoffs. Each month is a fresh decision, linked to earlier months only through the observed state.

Transitions, needed only for the forecast: $a' = \min(a^d + 1, \bar{A})$ after any action that leaves a car (so $a' = 1$ after B or R); after S, $n' = 0$.

Forecast by (3). On household–month data this is a discrete-time hazard model with competing outcomes (purchase, sale). It is the predictive baseline every later version has to beat.

What it gives. First purchases, replacements and exits from car ownership, from four parameters.

What RC means here. In a static model, buying earns $g - RC$ for one month only. Read literally, no household would pay a car's full price for one month of use. So in Versions 1 and 2, RC is not the price of a car. It is a fitted threshold that sets the monthly purchase rate. It becomes a cost spread over the car's future use only in the dynamic versions.

What it cannot do. Money does not enter, so it cannot say who can afford to buy. For a household without a car, the purchase probability $\Lambda(g - RC)$ is the same for everyone and does not change over time. The holding-time effect η mixes the rising cost of an old car with the option value of waiting.

Version 2: static, with Ability

State $x = (n, a, \ell)$, where $\ell \in \{1, 2, 3\}$ is a discrete liquidity level (low, medium, high). **Menu** unchanged.

$$U_2(x, d) = u(x, d) - \phi \, \mathbf{1}\{\ell = 1\} \, \mathbf{1}\{d \in \{\mathsf{B}, \mathsf{R}\}\}, \qquad \phi > 0. \tag{5}$$

ϕ is the cost of buying while illiquid: a financing premium or forgone consumption.

Transitions. Liquidity follows a Markov chain $\Pi(\ell' \mid \ell)$. A purchase lowers it one level first, with a floor at 1. A sale raises it one level first, with a ceiling at 3.[1] The forecast needs these even though the household is myopic.

What it adds. The first link to Ability. Two households with the same car can now have different purchase probabilities.

What it cannot do. A static household does not see that paying cash drains future liquidity, and it cannot wait on purpose for liquidity to improve.

Version 3: dynamic (forward-looking)

Same state, menu and flow utility as Version 2, with transitions T_d as described there. The household now values the future:

$$v(x, d) = U_2(x, d) + \beta \sum_{x'} \bar{V}(x') \, T_d(x' \mid x), \qquad 0 < \beta < 1, \tag{6}$$

$$\bar{V}(x) = \log \sum_{b \in \mathcal{M}(x)} e^{v(x,b)} + \gamma, \tag{7}$$

with γ Euler's constant. Choice probabilities are (1) with v in place of U. For example, at $n = 1$,

$$v(x, \mathsf{R}) - v(x, \mathsf{N}) = \underbrace{\eta a - RC - \phi \, \mathbf{1}\{\ell = 1\}}_{\text{this month}} + \beta \underbrace{\Big(\mathbb{E}[\bar{V}(x') \mid x, \mathsf{R}] - \mathbb{E}[\bar{V}(x') \mid x, \mathsf{N}] \Big)}_{\text{future: new car, less liquidity}}. \tag{8}$$

What it adds. *Dynamic* here means that today's action changes the future state, and the household takes that into account. It does not mean the menu grows. The second term of (8) is where the value of waiting appears. Its sign is not fixed: when it is negative, part of the future is better served by waiting, and when it is positive, buying now also improves the future. A household with medium liquidity may wait for liquidity to rise rather than buy now and drop to $\ell = 1$. A household with an old car may hold on because it expects to sell later.

Forecast by (3), with the dynamic choice probabilities.

[1] The effect of a sale on liquidity is a working assumption, open for discussion (Section 4).

Nesting. At $\beta = 0$, Version 3 is exactly Version 2. The nesting is algebraic. The parameters do not keep their meaning: RC is a one-month threshold at $\beta = 0$ and a cost spread over the car's future use at $\beta > 0$, so estimates of RC from the two versions are not comparable.

What it cannot do. Every household has the same purchase cost, so a household that keeps its car for a long time has no explanation other than its state.

Remark 1 (What the dynamics buy, and what they do not). The choice probabilities at each observed state are identified directly from data. A sufficiently flexible static model can reproduce the same probabilities as Version 3. Magnac and Thesmar (2002) show that β and the flow utilities are not identified without further restrictions. Their result is nonparametric. With the parametric flow utility used here, which has four to six parameters, β may be identified, but only by the functional form. That is weak identification, and in practice β would be fixed rather than estimated. So, inside the observed environment, Version 3 need not forecast better than a flexible Version 2. What it adds is the ability to answer questions the data do not show directly: a change in the purchase cost, for example, or behavior at states the data rarely reach. Whether we need that is question Q1 in Section 4.

Version 4: persistent unobserved types

Same state and menu. Each household has a fixed type $\tau \in \{1, \ldots, G\}$, known to it but not to us, and the purchase cost differs by type, RC_τ. The type probabilities depend on the first observed state, $\pi_\tau(x_{i1})$. This corrects for the fact that households first seen with a young car are more likely to be frequent buyers (Heckman's initial-conditions problem).

The state does not grow. Each month we observe the household's choice, and that updates our belief about its type:

$$w_{i\tau,t+1} = \frac{w_{i\tau t}\, \mathbb{P}_\tau(d_{it} \mid x_{it})}{\sum_s w_{ist}\, \mathbb{P}_s(d_{it} \mid x_{it})}, \qquad w_{i\tau 1} = \pi_\tau(x_{i1}). \tag{9}$$

The whole history enters through $G-1$ numbers, updated recursively, so the dimension stays fixed. The forecast mixes over types: $F_i(t, H) = \sum_\tau w_{i\tau,t+1} F_i^\tau(t, H)$, with each F^τ computed from (3).

Teaching example (invented numbers). Two types with equal prior weight, and constant monthly purchase probabilities of 0.05 and 0.01, ignoring the state. After 20 months without a purchase, the weight on the first type falls from 0.50 to 0.30. The forecast monthly probability falls from 0.030 to 0.022, even though neither type's own probability has changed. A purchase rate that falls with holding time can come from differences between households, not from a change within any household.

What it adds. It separates "this household keeps cars long" from "this car is still young."

What it cannot do. Types are fixed for life, so a change in circumstances (a new job, a child) cannot be represented; that is Version 8. With rare purchases, the type shares may also be estimated imprecisely.

Who learns. Here we learn about the household, but the household does not learn about itself. A household learning its own switching inconvenience would be a different model, with a belief added to the state.

Version 5: which car

State $x = (n, a, q, \ell)$, where $q \in \{\text{used}, \text{new}\}$ is the tier of the car held. Tier enters the state because used and new cars age differently.

Menu. At $n = 0$: $\{\mathsf{N}, \mathsf{B}_{\text{used}}, \mathsf{B}_{\text{new}}\}$. At $n = 1$: $\{\mathsf{N}, \mathsf{R}_{\text{used}}, \mathsf{R}_{\text{new}}, \mathsf{S}\}$.

Utility. The flow (4) with tier-specific parameters:

$$u(x,d) = n^d\big(g_{q^d} - \eta_{q^d} a^d\big) - RC_j\,\mathbf{1}\{d \in \{\mathsf{B}_j, \mathsf{R}_j\}\} - \kappa_S\,\mathbf{1}\{d = \mathsf{S}\}, \tag{10}$$

plus the liquidity term of Version 2, now tier-specific as ϕ_j with $\phi_{\text{new}} > \phi_{\text{used}}$. The purchase probability sums over tiers, and (3) is unchanged.

Transitions. The outlay now differs by tier. A purchase of tier j lowers liquidity by Δ_j levels before the Markov step, with a floor at 1: $\Delta_{\text{used}} = 1$ and $\Delta_{\text{new}} = 2$. This replaces the single one-level drop of Version 2.

What it adds. The size of the outlay, through Δ_j and ϕ_j, which is what links a purchase to a loan amount. Low liquidity can now push a household to a used car rather than stop the purchase: at $\ell = 1$ a new car costs ϕ_{new} now, and in the dynamic versions it also leaves less liquidity for later.

What it cannot do. Every purchase is implicitly paid in cash; financing comes in Version 6.

Caveat. Plain multinomial logit treats the two purchase options as no closer to each other than to doing nothing (independence of irrelevant alternatives). A nested logit, with the purchase options in one nest, is the obvious check.

Version 6: financing

Menu. Each purchase action comes in two forms: $\mathsf{B}_{j,f}$ and $\mathsf{R}_{j,f}$, with $f \in \{\text{cash}, \text{loan}\}$.

State adds $b \in \{0, \ldots, \bar{b}\}$, the months left on the current auto loan; $b = 0$ means no loan. Contract terms are fixed by tier, not chosen.

Rules. Liquidity changes are in levels. The outlay Δ_j is the one from Version 5. Unlike Versions 2–5, an action is available only if it leaves liquidity at level 1 or above, so the floor never binds and no purchase is partly free.

- Cash lowers liquidity by Δ_j. With three levels, a used car needs $\ell \geq 2$ and a new car needs $\ell = 3$.
- A loan sets $b' = \bar{b}$ and lowers liquidity by $\Delta_j - 1$ levels, the down payment: nothing for a used car, one level for a new car, which therefore needs $\ell \geq 2$.
- While $b > 0$, liquidity moves by a chain $\Pi_L(\ell' \mid \ell)$ that is shifted down relative to Π, because the monthly payment drains cash. Each month with a loan also costs κ_L in flow utility, the interest. Then b falls by one.
- Selling or replacing a car with $b > 0$ pays off the loan: liquidity falls one more level. So a sale of a financed car leaves liquidity unchanged, and replacing a financed car with a new car for cash is not available.
- The loan option exists only if the lender approves it. Let $A_{it} \in \{0, 1\}$ be approval, with $\mathbb{P}(A_{it} = 1 \mid x_{it})$. Creditworthiness enters here, and it remains deferred.

Choice probabilities. Approval changes the menu. Let $\mathcal{M}(x, A)$ be the menu with the loan options removed when $A = 0$. Then

$$\mathbb{P}(d \mid x) = \sum_{A \in \{0,1\}} \mathbb{P}(A \mid x)\, \frac{\exp v(x,d)\,\mathbf{1}\{d \in \mathcal{M}(x,A)\}}{\sum_{b \in \mathcal{M}(x,A)} \exp v(x,b)}. \tag{11}$$

In the continuation value, the household must also form expectations over next month's approval.

What it adds. It separates *buying a car* from *taking an auto loan*, and the loan is the event in the original exercise. The loan forecast $F_i^{\text{loan}}(t, H)$ follows from (3) once the cash purchases are added to the non-purchase set $\mathcal{N}(x)$.

What it cannot do. Approval is taken as given rather than modeled, and the household cannot choose among lenders or terms. The loan rate does not appear in the model as a variable.

It enters only through κ_L and Π_L, both fixed by tier, and with fixed terms the data show no variation in it. So a rate counterfactual is not identified in this version.

Approval and preference. If denied applications are not observed, a household with no loan may have been refused, or may not have wanted one. Purchase data alone cannot separate $\mathbb{P}(A \mid x)$ from the preference for a loan. That needs application and denial records (question Q6) or an exclusion restriction: a variable that moves approval but not preference.

Version 7: second cars

This version raises the cap to $\bar{n} = 2$. It builds on Version 4, not Version 6: tier and financing are left out to keep the state small. Combining the branches would give the state $(n, a_1, q_1, a_2, q_2, \ell, b)$, of dimension 7.

State $x = (n, a_1, a_2, \ell)$, where $a_1 \geq a_2$ are the holding times of the two cars; "older" below means the car held longer. Empty slots are coded as such, so the dimension is fixed.

Menu.

- At $n = 0$: $\{\mathsf{N}, \mathsf{B}\}$.
- At $n = 1$: $\{\mathsf{N}, \mathsf{R}, \mathsf{S}, \mathsf{A}\}$.
- At $n = 2$: $\{\mathsf{N}, \mathsf{R}, \mathsf{S}\}$, where R and S act on the older car.

A purchase is any B, R or A.

Utility. $u(x, d) = \sum_{k=1}^{n^d} \big(g_k - \eta\, a_k^d\big) - RC\, \mathbf{1}\{d \in \{\mathsf{B}, \mathsf{R}, \mathsf{A}\}\} - \kappa_S\, \mathbf{1}\{d = \mathsf{S}\}$. Here g_2, the value of a second car, may be smaller than g_1. Choice probabilities are (1) with the dynamic values over the menu at each n.

Forecast by (3), with $\mathcal{N}(x) = \{\mathsf{N}, \mathsf{S}\}$ whenever $n \geq 1$.

What it adds. Additions, which the proposal counts as purchases. Versions 1–6 cannot see them.

What it cannot do. Households with three or more cars are outside the model. Whether to drop them or code them at the cap is a real-data question.

Two separate extensions. As you pointed out, "more kinds of car, quantity one" (Version 5) and "more than one car" are different extensions. This version is still a discrete choice: the number of cars is a bounded integer, changed one transaction at a time. Buying two cars in the same month would be a further version.

Version 8: latent need

Add a hidden state $z_{it} \in \{0, 1\}$ (normal, high need) to any earlier version. It follows a Markov chain $\mathbb{P}(z' \mid z)$. High need makes each missing car, relative to the cap, costly:

$$u(x, z, d) = u(x, d) - \zeta\, z\, (\bar{n} - n^d), \qquad \zeta > 0. \tag{12}$$

At $\bar{n} = 1$, need pushes a household without a car to buy one and discourages selling. At $\bar{n} = 2$, it pushes both a first purchase and the addition of a second car. An indicator $\mathbf{1}\{n^d < \bar{n}\}$ in place of the shortfall would fail here: at $\bar{n} = 2$ it penalizes zero and one car equally, so need would have no effect on buying a first car.

The household knows z. We see it only through signals w_{it} in transactions, such as moving costs, childcare or repair spending, with emission law $f(w \mid z)$. Our belief $p_{it} = \mathbb{P}(z_{it} = 1 \mid$ history up to $t - 1)$ is again one number. Unlike the types in (9), z changes over time, so the update has two steps. First, correct with this month's signal and choice:

$$\tilde{p}_{it} = \frac{p_{it}\, f(w_{it} \mid 1)\, \mathbb{P}(d_{it} \mid x_{it}, z = 1)}{p_{it}\, f(w_{it} \mid 1)\, \mathbb{P}(d_{it} \mid x_{it}, z = 1) + (1 - p_{it})\, f(w_{it} \mid 0)\, \mathbb{P}(d_{it} \mid x_{it}, z = 0)}. \tag{13}$$

Then predict next month's need through the transition:

$$p_{i,t+1} = \tilde{p}_{it}\mathbb{P}(z' = 1 \mid z = 1) + (1 - \tilde{p}_{it})\mathbb{P}(z' = 1 \mid z = 0). \tag{14}$$

This is the standard hidden-Markov filter. The forecast runs (3) on the joint state (x, z), starting from $p_{i,t+1}$.

What it adds. This is where life events belong: a new job, a move, a baby. They change the state through z; they do not enlarge the menu.

Cost. It adds a transition and an emission equation. Identification requires that the signals affect the choice only through z. That is a strong restriction, and none of the papers reviewed establishes it for bank transaction data.

Which versions the question needs

A provisional view, for discussion:
- Versions 1–4 are the core for the purchase probability.
- Version 6 is needed for the auto-loan event.
- Version 7 is needed for the full purchase definition, since it adds additions.
- Version 8 is the most speculative and the hardest to identify.

Each step should be kept only if it improves the calibrated forecast of $F(t, H)$ on held-out households over the previous version, or if it is needed for a stated counterfactual.

3 Choice of decision period

Four quantities to keep separate.
- the decision period (how often the model lets the household choose);
- the frequency at which the data record the state;
- the precision of the purchase date;
- the forecast horizon H in calendar months.

They need not be equal. Comparisons across periods should hold the target fixed at 6 and 12 calendar months, because accuracy per row is not comparable across period lengths.

Imbalance arithmetic. Using the illustration from our discussion (10 purchases over 60 years of driving), Table 2 shows what changes with the period length.

	Monthly	Quarterly	Semiannual	Annual
Rows per household	720	240	120	60
Share of rows with a purchase	1.4%	4.2%	8.3%	16.7%
Purchases	10	10	10	10
Periods to $H = 6$ / $H = 12$ months	6 / 12	2 / 4	1 / 2	– / 1
Purchase periods with ≥ 2 purchases†	0.7%	2.1%	4.1%	8.1%
Discount factor per period‡	0.9957	0.9873	0.9747	0.95
Value-iteration steps to 10^{-8}	4,310	1,437	719	360

Table 2: Illustrative figures, not data. †Poisson approximation with a constant rate; it overstates the case for replacements, since the rate falls right after a purchase. ‡Assuming an annual factor of 0.95.

What coarsening changes and what it does not.

1. **It does not create purchases.** Only non-purchase rows disappear. In a simple special case, King and Zeng (2001, eq. 12, p. 147) approximate the bias in the logit intercept as $(\bar{\pi} - 0.5)/[n\bar{\pi}(1-\bar{\pi})]$, which for rare events is about $-1/(2n\bar{\pi})$. It shrinks with the expected number of events, $n\bar{\pi}$, and coarsening the period leaves that number unchanged. So a coarser period makes the imbalance look smaller but does not reduce this bias on its own.
2. **It reduces computation.** There are fewer rows, and the discount factor per period is further from 1. Value iteration needs about a third of the monthly number of steps at quarterly frequency.
3. **It loses timing.** The state is measured at the start of the period. A tax refund in the first month of a quarter followed by a purchase in the second falls inside one period. Measuring at the end instead would leak the purchase into the state, since paying cash lowers the balance.
4. **It strains the one-transaction rule.** With selling now in the menu, a sale and a purchase in the same period must be coded as a replacement or forced into one action. The longer the period, the more often this happens.
5. **It can make the forecast trivial.** At semiannual frequency, $H = 6$ is a single period, so the dynamics play no role in that forecast.

Working choice. Monthly for Versions 1 and 2. Static versions are cheap at any number of rows, monthly keeps the timing signal, and it matches the frequency of bank data. The period for Versions 3 and above is open. If it changes, Version 2 should be refitted at the new period first. Otherwise the step from Version 2 to Version 3 changes two things at once, and the comparison cannot separate them. I have not yet compared frequencies on real data.

4 Working choices and questions for you

Decided for this draft.

- **Actions:** buy, replace, sell, do nothing, and add once the cap allows it.
- **Number of cars:** raised one step at a time.
- **Purchase:** a first car, a replacement, or an addition.
- **Period:** monthly for the static versions.

Working assumptions I would like to check with you.

- The flow utility is linear in money (risk-neutral), consistent with deferring risk attitude.
- Income is left out of the state. With linear utility and no saving decision, it adds the same amount to every action and cancels from the choice. It can matter only through how liquidity evolves.
- Liquidity has three levels. A purchase lowers it one level (by tier from Version 5 on) and a sale raises it one level.
- There are two persistent household types, with type probabilities that depend on the first observed state.

Questions.

Q1. When the bank uses this, is the output a ranked list of customers likely to buy in 6 or 12 months under current conditions? Or must it also predict how that changes under some change (a cheaper car, a different loan offer), and if so, which change? The answer decides whether Versions 3 and above are needed (Remark 1).

Q2. Is a cap of one car acceptable for Versions 1–6, with second cars added in Version 7?

Q3. How should liquidity be measured in bank data: the checking balance, or cash across all accounts relative to monthly spending?

Q4. In bank data, how would we know that a household has no car? No auto loan does not mean no car.

Q5. Which state variables can actually be observed in the bank setting, and how often: car age and tier, liquidity, loan status, number of cars?

Q6. Does the bank see loan applications and denials, or only approved loans? Without denials, Version 6 cannot separate "was refused" from "did not want a loan."

Q7. Near the end of our discussion you said something like "...should be a hundred, this is way too large to start with." The recording is unclear. Did that refer to the size of the state space, the menu, or the sample?